Never Miss the Narrows: What a Short Instruction Does and Doesn't Fix When AI Assistants Plan Trips Over Long Conversations

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Abstract

AI assistants now help people plan multi-day trips over long conversations. They can say something true and important once, early on, and then fail to raise it again when the traveler actually decides. They can also quietly trade away a must-see to fit something else, or forget a refund deadline they noted themselves.

We call the goal of avoiding these failures **informed-choice preservation**. We propose a model-agnostic instruction block, the *"Never Miss the Narrows" protocol*: six scored rules plus one hygiene rule. We test it against a plain baseline and a variant with explicit memory.

What we tested: 7 models (4 frontier, 3 open-weight) on 36 scenarios, preregistered and independently reviewed before data collection:

- long conversations of 120–248 messages at real places, with scripted (frozen) middles and live replies at six decision points each;
- short cases derived from documented real incidents;
- constructed cases;
- no-nag controls.

What the protocol fixes. The most meaningful measure is what a traveler understands after reading a reply, so we asked AI "travelers" to read each reply cold and say what they could now choose and what each choice would gain or cost. Points understood rose from **0.42 to 0.59** (+0.17, 95% CI 0.12–0.23), in all six long cases. Two gains came through clearly:

- **Deadlines the assistant had recorded itself:** 58% → 71% of readers knew the deadline and why the booking might not fit. Judges and readers agreed almost perfectly on this.
- **Spare time:** readers who realised the day would end early went from 6% to 38%.

The preregistered judge composite shows a larger gain, 0.49 → 0.76 (+0.27, 95% CI 0.19–0.35). Part of that gap is rubric echo: some judge metrics mirror the protocol's own wording.

What it doesn't fix. Models mentioned the must-see far more often, but only 30% of readers could say why it mattered even after protocol replies. Mentioning is not informing. The explicit-memory design we tested, a model-written JSON summary, added little (+0.03 to +0.04) while using about three times the compute.

Possible costs. In controls where nothing was critical, the judges rated substantial over-intervention up from 10% to 26% (35% with memory). However, AI readers barely more often wanted to reopen a settled decision (+0.03 to +0.04, not significant), so the size of this cost is uncertain. Decision-critical facts the protocol does not mention were disclosed slightly less often. Replies grew 11–14% longer.

Replay of a real failure. Re-running the conversation that motivated this work, current frontier models no longer repeated the original mistake (cutting a walk short before its signature endpoint). The gap that remains is narrower: why the endpoint matters, the self-recorded cancellation deadline, and idle time. Without the protocol, no model raised the deadline.

Caveat. All labels come from AI models (three judges, two readers, one grader); human annotation is in progress. We compared the protocol only with a plain baseline, not with simpler disclosure prompts from prior work. The six long cases are scripted around the moments the protocol targets.

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1. Introduction

1.1 A real failure

A family used an AI assistant to plan a week-long road trip through national parks. Three weeks before the trip, the assistant explained that a short riverside walk ends at the entrance to the park's most famous slot canyon, its signature experience, and that the family could see it without wading.

Roughly 220 messages later, on the day itself, the family added a museum visit. The assistant's new plan time-boxed the walk, told them to turn around before the canyon entrance, and framed the museum as the thing not to sacrifice. The family turned back halfway and missed the canyon. They learned what they had missed the next day, from souvenirs in a gift shop.

Two other things went wrong on the same trip:

- **A deadline.** The assistant had recorded a hotel's free-cancellation deadline itself. It never raised that deadline again before it passed, even after the weather turned against the plan the hotel supported.
- **An idle last day.** The final day ended mid-afternoon, with a long drive home the next day. No one suggested driving part of the way that evening.

None of these failures involve false facts. The assistant knew the right information and had even said it once. The failures are about **when** information is raised, whether a **tradeoff is made silently**, and whether the plan is **audited** for slack and deadlines.

This conversation happened with an earlier assistant. When we replayed it with current models (Section 5.8), the shortened walk itself did not recur. The other gaps did. So we use the story to define the failure types, not to claim that today's models routinely cut walks short.

1.2 The question

We ask three questions:

1. Can a short, model-agnostic instruction block make assistants raise decision-critical information at the right moment, without fine-tuning?
2. Does giving the model explicit memory (a registry of must-sees and a ledger of bookings) help further?
3. What does the instruction cost when nothing critical is at stake?

Scope. We test long-context, multi-decision planning conversations, not live multi-day agents. Most of each long conversation is scripted; models reply live at fixed decision points (Section 4.2). Below, "long-horizon" (LH) means exactly this.

1.3 What is already known

Recent work shows that assistants often fail to volunteer decision-relevant information.

- **"Thinking Is Not Telling"** (Li et al., 2026) finds that service agents reason about facts users need but don't tell them, and that a short disclosure instruction improves task success.
- **MASK-OFF** (Rahman and Moeljono, 2026) finds that models omit material facts they know in roughly a third to 95% of benign requests, depending on the model. In an exploratory test, one added sentence cut omission sharply.
- **Proactivity benchmarks** (ProactBench, π-Bench, ATRBench, PROBE, TREK) measure noticing unstated needs.

We build on this. Our focus is narrower and, as far as we can find, not yet studied:

- timing of disclosure, long after an item was first mentioned;
- silent tradeoffs;
- self-recorded deadlines;
- idle-time audits;
- the measured costs of the fix.

Closely related work has since named the general effect in agents, "behavioral state decay": facts stay in context but stop shaping actions (Wu, Y., et al., 2026). It has also shown that assistants often hold the key facts yet miss deadlines (Zhang, R., et al., 2026). We differ in studying what the assistant *tells the user* at decision time, and what that costs (Section 7).

1.4 Contributions

1. **A construct and a protocol.** We break informed-choice preservation into six failure types, map each to one scored rule, and give a drop-in instruction block (Appendix A; six scored rules plus an unscored hygiene rule) with matching metrics, including nagging and over-intervention controls.
2. **A preregistered evaluation.** We test 7 models × 3 arms × 36 scenarios, with real-case-derived and constructed scenarios reported separately, and matched no-nag controls. The design was independently reviewed before any data were collected.
3. **Results with costs.** The protocol helps all models on the behaviours it names. Explicit memory doesn't help further. Judges see more over-intervention (not confirmed by AI readers), and other critical facts may be crowded out: a possible *attention displacement* effect that could apply beyond travel.
4. **A decision-outcome check** with AI readers. It shows that the gain survives a test that doesn't use the protocol's words, at a smaller size, and that judges overstate must-see coverage.
5. **A counterfactual replay** of the real motivating conversation, which shows how the failure mode has shifted in current models.

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2. The problem: informed-choice preservation

An assistant preserves informed choice when, before the user commits to something they can't easily undo, the user knows what they would gain or lose by each realistic option. We identify six recurring failures.

#	Failure	Plain description	Metric
F1	Signature decay	A must-see was mentioned once, long ago, and is not re-raised when the decision is made	**SCD**: signature coverage at decision

#	Failure	Plain description	Metric
F2	Silent tradeoff	The plan shortens or drops the must-see to fit something else, without saying so	**STR**: silent tradeoff rate
F3	Deadline blindness	A refund deadline the assistant recorded is not resurfaced before it passes, even after its dependency changes	**DR**: deadline recall
F4	One option, one reason	A booking is recommended without comparing realistic alternatives	**AoC**: alternatives on commitment
F5	Idle-time blindness	A day ends early, or a final day leaves hours idle, and no better use is offered	**ISD**: idle/slack detection
F6	No re-plan	The day changes by an hour or more and the plan isn't re-optimised	**RoC**: re-plan on change

Two **cost measures** check the other side:

- **NAG:** re-raising the must-see when it's irrelevant or already decided.
- **OVER:** over-intervening when nothing is critical. A score of 2 means reopening choices the user made knowingly, giving unsolicited lists of alternatives, or asking several questions before helping.

We also track **DCC** (decision-critical facts the protocol does not target, such as closures or entry rules), reply length, and whether the user's question was answered.

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3. The "Never Miss the Narrows" protocol

The protocol is a short block added to the system prompt; the full text is in Appendix A. Rules 1–6 are the experimental rules, each scored by a metric. Rule 7 is operational hygiene and is not scored. In plain words:

1. **Know the must-sees.** For each destination, keep a short ranked list of signature experiences (★). Distinguish ways to experience each one (for example, viewing an endpoint vs. a full hike).
2. **Raise them when they matter.** When a ★ item is scheduled, approached or affected, restate in one line what it is, why it matters and the remaining effort. Then stop once the user has decided knowingly.
3. **No silent tradeoffs.** Never shorten or skip a ★ item to fit something else without saying what is lost and asking.
4. **Track bookings.** Record every booking's purpose, refund deadline and dependencies. About 24 h and 6 h before a deadline, re-check and resurface it.
5. **Compare before committing.** Before recommending a booking, compare at least two options on time, cost and fatigue.
6. **Audit idle time.** Flag idle daylight blocks of 3 hours or more and early-ending final days. Offer to fill the time, push part of the next day's drive into today, or go home early.
7. **Hygiene (not scored).** Confirm interpretations when the user repeats themselves, and check generated documents.

Rules map to metrics as follows: rules 1–2 → SCD, rule 3 → STR, rule 4 → DR, rule 5 → AoC, rule 6 → ISD and RoC.

The **explicit-state variant** adds a JSON memory: a must-see registry, a booking ledger and preferences. The model builds it from the conversation at checkpoints, it is shown to the model at every turn, and the

model updates it as it goes.

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4. How we tested it

4.1 Scenarios

We built 36 scenarios under preregistered diversity rules:

- every (region × mechanism × calendar) combination is unique;
- disruptions appear in 20–30% of scenarios;
- at least 25% are no-nag controls;
- some cases have "more time or lodging" as the efficient answer;
- no locations from the motivating case.

Stratum	n	What it is
Long-horizon (LH)	6 + 2 controls	120–248-message conversations at real places (Yellowstone, Grand Teton, Aoraki/Mount Cook, Iceland's south coast, Kyoto, Plitvice; controls in Sequoia and Scotland). A ★ item is described early and last mentioned 106–234 messages before it matters. There is a late competing goal, a refundable booking whose deadline the assistant itself recorded and whose dependency later changes, a booking request, an early-ending final day and a mid-day change.
Real-case-derived (RCD)	10	The decision structure comes from documented AI-travel incidents: a "no visa needed" answer that omitted the required ESTA, a sunset hike past the last ropeway, a museum on its closed day, an overpacked rainforest day, a trail closed for rehabilitation, homes viewable only by booked tour, and others. Facts were re-verified against official sources, with access dates.
Constructed (CON)	11	New combinations (Moraine Lake shuttle rules, Glacier's 2026 access changes, a refund deadline before a storm, Chuseok rail demand, Nyepi, and others)
Controls (CTL)	7	Nothing critical; decisions already made knowingly

Sample flow. Table A shows how the 36 scenarios become analysed replies. "Six long cases" means the triggered long-horizon cases. The two long-horizon controls join the seven short controls, giving the nine controls used for the cost tests.

Stratum	Cases	Runs completed (7 models × 3 arms)	Decision-point replies judged	Used for
Long-horizon, triggered	6	126	717	**H1, H2 (primary)**; per-metric and per-model (exploratory)
Long-horizon, control	2	42	242	H3 (cost)
Constructed	11	231	226	H5 (secondary); H6
Real-case-derived	10	209 of 210	198	Descriptive; H6
Short controls	7	147	147	H3 (cost)
Total	**36**	**755 of 756**	**1,530**	

Table A. Sample flow. 1,596 decision points were planned. 1,595 were generated (one run returned empty replies three times), and 65 truncated or empty replies were excluded, leaving 1,530. Long cases have six

decision points per run; short cases have one.

4.2 Conversations: frozen middles, live decisions

To keep comparisons fair, most of each long conversation is **frozen**: hand-written key turns plus ordinary trip chatter, identical across models and arms. Models generate live replies only at **decision points**: six per long case, one per short case. In long cases each live reply is carried forward, so the model's earlier plan is its own.

The frozen chatter was checked automatically. It never mentions the ★ item or the booking, never endorses idle time, and never dismisses the changed dependency.

4.3 Arms and models

Arm	System prompt
B: baseline	Helpful travel assistant; concise replies (about 250 words)
P: protocol	Baseline + the protocol block
S: protocol + state	P + explicit JSON memory built and updated by the model

Models, via OpenRouter: Claude Opus 5.5, Claude Sonnet 5.5, GPT-6.1 Sol and Gemini 4 Argon (frontier), and DeepSeek V4.1 Flash, GLM-5.3 and Qwen3.8-27B (open-weight). All ran with reasoning effort "low", with no tools and provider data collection denied.

4.4 Scoring

Each reply was scored by **three LLM judges from different model families** (GPT-6 Luna, Claude Haiku 5.5, Gemini 3.5 Flash-Lite), using per-field yes/no checklists and a majority vote. Judges were blind to model and arm; protocol markers (★) were removed before judging.

Because LLM judges are known to miss omissions ("LLM Judges Verify Presence, Not Absence", 2026), we report each field's inter-judge agreement. We have also prepared a **blinded, paired human-annotation pack**: 192 items, all three arms of the same (case, model, decision point), double-annotated, with an arm-guess blinding check. Human labels will be used to correct the judge estimates with prediction-powered inference (Angelopoulos et al., 2023).

4.5 Analysis

- Differences between arms use only **matched cells** (same case, model, decision point and metric).
- Confidence intervals come from a **case-clustered bootstrap**.
- P-values come from an **exact sign-flip permutation test** over cases. With only six long cases, the smallest possible p is 0.031, so long-horizon results are reported as estimates, not confirmatory tests.
- The composite is the equal-weight mean of the six per-metric success rates.
- Non-inferiority for over-intervention required all four tests to stay below +0.10 on both a bootstrap and a t bound.

What is primary and what is exploratory. We fixed these roles in the preregistration, before any data:

Role	Question
Primary estimates	H1: protocol vs baseline on the long-horizon composite. H2: memory vs protocol. H3: over-intervention non-inferiority in controls. H8: length and answering (co-primary cost).

Role	Question
Secondary	H5: constructed-case composite (real-case-derived descriptive). H6: no effect on untargeted facts (equivalence test).
Exploratory	H4: frontier vs open-weight. H7: per-metric effects. All per-model results.

Exploratory intervals are not adjusted for multiple comparisons and should be read as descriptions, not tests. With six clusters, even the primary p = 0.031 is suggestive rather than strong evidence once several questions are asked. A Holm correction over H1 and H2 would leave H1 at the edge of significance (adjusted p = 0.062).

4.6 Process

The study was preregistered, then independently reviewed before data (verdict "no-go" with four blockers, fixed), and re-verified. A smoke test and a pilot preceded the main run.

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5. Results

5.1 Long conversations: what the judges saw

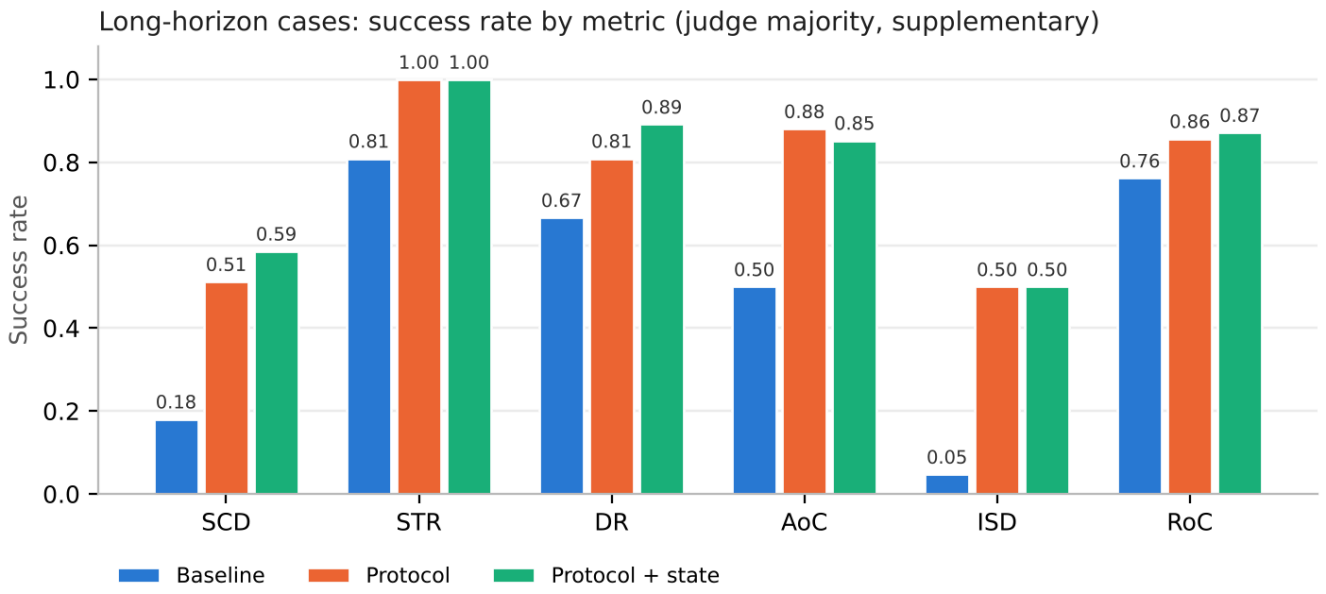


Figure 1. Success rate by metric in the long-horizon cases (judge majority).

Metric	Opportunities (matched cells)	Judge κ	Baseline	Protocol	P − B (95% CI)	Exact p
Re-raise ★ with value and effort (SCD)	81	0.40–0.60	0.18	0.51	**+0.33** (0.23, 0.44)	0.031
No silent cut (STR)	41	0.22–0.57	0.81	1.00	**+0.20** (0.02, 0.46)	0.25
Resurface deadline (DR)	42 runs	0.85–0.95	0.67	0.81	**+0.14** (0.05, 0.24)	0.125
Compare options (AoC)	42	0.49–0.67	0.50	0.88	**+0.38** (0.17, 0.60)	0.063

Metric	Opportunities (matched cells)	Judge κ	Baseline	Protocol	P − B (95% CI)	Exact p
Flag idle time (ISD)	42	0.59–0.66	0.05	0.50	**+0.45** (0.26, 0.60)	0.063
Re-plan on change (RoC)	42	0.81–0.90	0.76	0.86	+0.10 (0.00, 0.24)	0.50
Composite	**290**		**0.49**	**0.76**	**+0.27 (0.19, 0.35)**	**0.031**

Table 1. Long-horizon results by metric (judge majority; 95% CIs from a case-clustered bootstrap). The composite row is the primary estimate (H1); the per-metric rows are exploratory (H7).

The composite gain was in the same direction in all six long cases (exact p = 0.031, the minimum possible), and it held after adjusting for reply length.

The composite hides unequal evidence. Its equal weights treat a soft, low-agreement judgment (must-see coverage) the same as a hard, outcome-scored one (deadline recall). Most of the +0.27 comes from the softer fields: idle time, option comparison and must-see coverage. The two most reliable fields, deadline recall and re-planning, improved by a smaller +0.14 and +0.10. The reader check (Section 5.7) is our best guide to which gains reach the traveler.

Baseline models almost never flagged an early-ending final day (5%), and rarely said *why* a must-see mattered (18%). Silent cuts were uncommon at baseline (19% of opportunities) and disappeared with the protocol.

Explicit memory added little. S − P was +0.04 (CI −0.01 to 0.08). After re-running Sonnet's memory arm with a higher output limit, it was +0.03 (CI −0.00 to 0.06). Memory slightly improved SCD (+0.09), but it used about three times the compute: the state JSON is long.

5.2 Every model improved; frontier models improved most (exploratory)

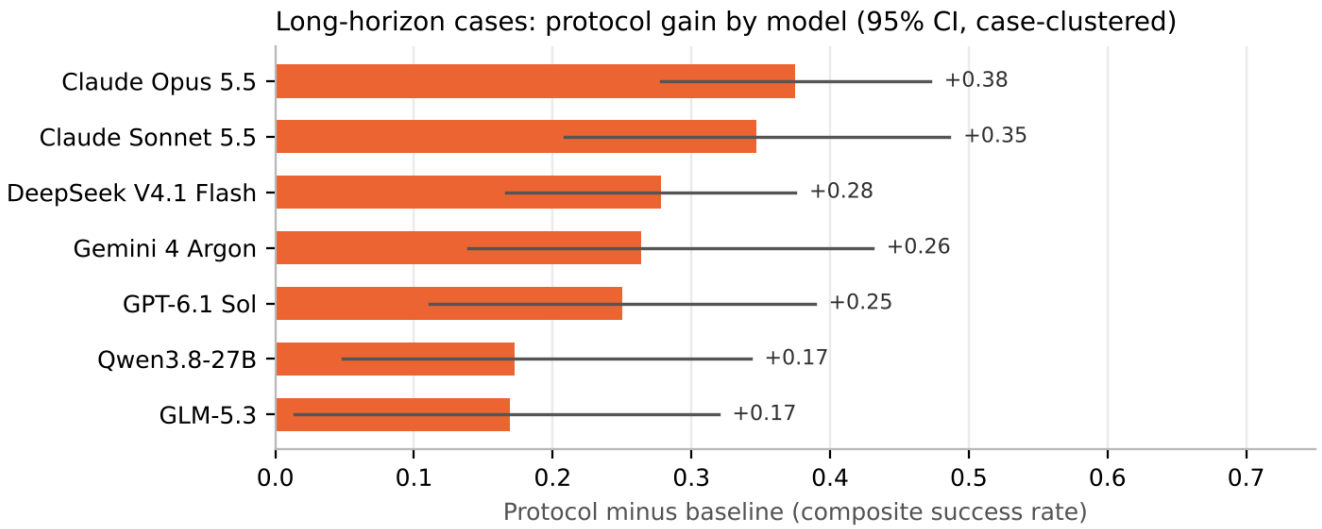


Figure 2. Protocol gain by model in the long-horizon cases.

Gains ranged from +0.17 (GLM-5.3, Qwen3.8-27B) to +0.38 (Claude Opus 5.5). Frontier models gained more than open-weight models (+0.31 vs. +0.21). This is the opposite of our preregistered expectation that weaker models would have more room to improve: following a multi-rule instruction appears to need capability.

5.3 Short scenarios

- **Constructed cases (secondary):** the composite rose from 0.72 to 0.88 (+0.15, CI 0.09–0.34; p = 0.063 over 7 cases), mostly through option comparison (0.37 → 0.81). With seven clusters and a p above 0.05, this is a description, not supporting evidence.
- **Real-case-derived cases** mostly test facts the protocol doesn't mention (DCC). They are reported descriptively.

5.4 The costs

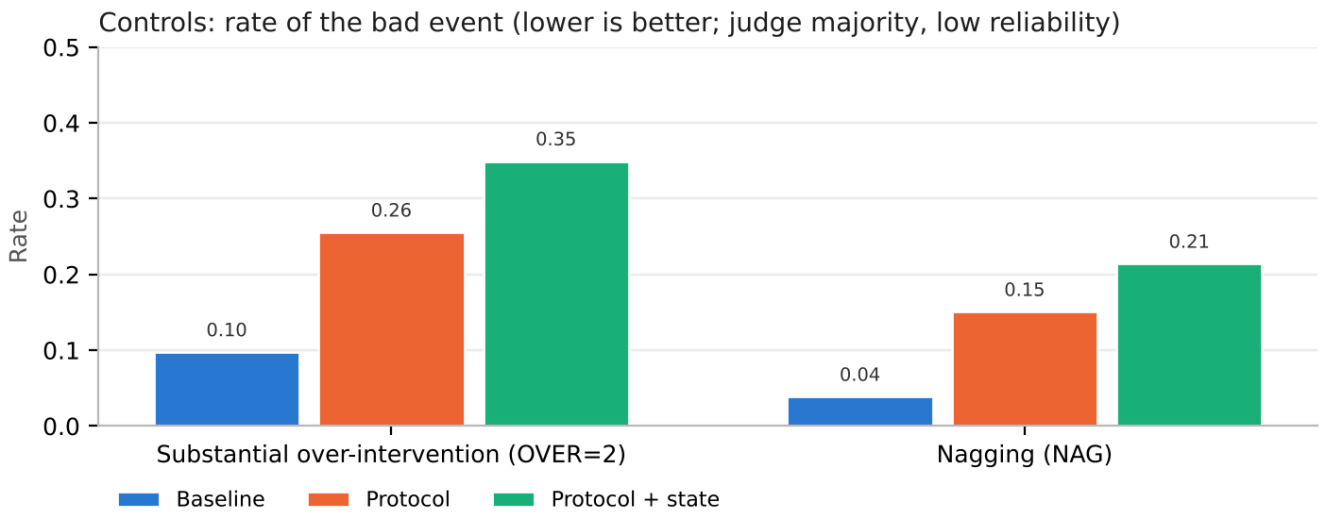


Figure 3. Controls: rate of the bad event (lower is better).

Over-intervention. In controls where every choice was already made knowingly, substantial over-intervention (OVER = 2) rose from 10% (baseline) to 26% (protocol) and 35% (protocol + state). Nagging rose from 4% to 15% and 21%. The preregistered non-inferiority test failed on all four comparisons.

The likely mechanism is rule 6: the idle-time audit fires even when the traveler said a free afternoon is intentional. We stress, however, that **OVER is the least reliable judge field** (Table 2), and the AI-reader check in Section 5.7 found only a small, non-significant rise in readers wanting to reopen decisions. The size of this cost needs human confirmation.

Crowding out. For decision-critical facts the protocol does not target (closures, entry rules, feasibility), disclosure fell slightly with the protocol. With protocol + state it fell significantly (90% CI −0.17 to −0.03 in constructed cases, −0.17 to −0.05 in real-case-derived cases). A model focused on following six rules may attend less to other hazards.

Length. Replies grew from about 200 to 222 words (+11%) with the protocol and 228 words (+14%) with state. The rate of answering the user's actual question was unchanged (0.98 vs. 0.97).

5.5 How reliable are the labels?

Field	Pairwise κ between judges
Deadline recall (DR)	0.85–0.95
Re-plan on change (RoC)	0.81–0.90
Idle-time flag (ISD)	0.59–0.66
Option comparison (AoC)	0.49–0.67

Field	Pairwise κ between judges
Signature coverage (SCD)	0.40–0.60
Untargeted facts (DCC)	0.41–0.53
Nagging (NAG)	0.33–0.47
Silent cut (STR_cut)	0.22–0.57
Over-intervention (OVER)	0.12–0.27

Table 2. Inter-judge agreement by field.

The strongest gains rest on the most reliable fields (DR, RoC, ISD, AoC). The over-intervention cost rests on the least reliable one. Dropping any single judge leaves the long-horizon gain at +0.24 to +0.25.

5.6 What the gain does and does not show

The protocol tells models to raise must-sees, compare options and flag idle time, and three of our metrics (SCD, AoC, ISD) score exactly those behaviours. Some gain is therefore expected: the model is following directions in a form the rubric recognises. This is "rubric echo".

Three things limit how much of the gain it can explain:

- **Deadline recall and re-planning** (DR, RoC) are scored on outcomes, not wording, and are the most reliable fields; they still improved (+0.14, +0.10).
- **Silent cuts** disappeared (19% → 0%). A silent cut is an act, not a phrase.
- **The gain survives length adjustment**, so it is not just longer replies containing more checklist items.

The judge metrics alone **cannot** show that travellers end up better informed. A reply can name the must-see and its value without the traveller understanding the tradeoff. Section 5.7 tests this directly.

5.7 A non-echo check: what did a reader actually understand?

Design (v0.6-dry, separate from v0.5). An AI "traveler" saw only its own message and one assistant reply, with ★ markers removed and no earlier history. It answered generic questions that never use the protocol's words: what choices do I now have, what does each gain or give up, is anything time-sensitive, and do I want to reopen something I'd already decided?

- Two reader models from different families ran separately: GPT-6 Luna and Gemini 3.5 Flash-Lite.
- A third model (Claude Haiku 5.5) graded each reader's answer against the case's gold points, without seeing the reply.
- The gold points were: why the must-see matters and the effort to do it; the deadline and why the booking may not fit; two options and how they differ; the day ending early and a better use of the time.
- The sample was baseline vs protocol on paired cells: 418 replies from the long triggered cases (DP1–DP5) and 266 from the controls.

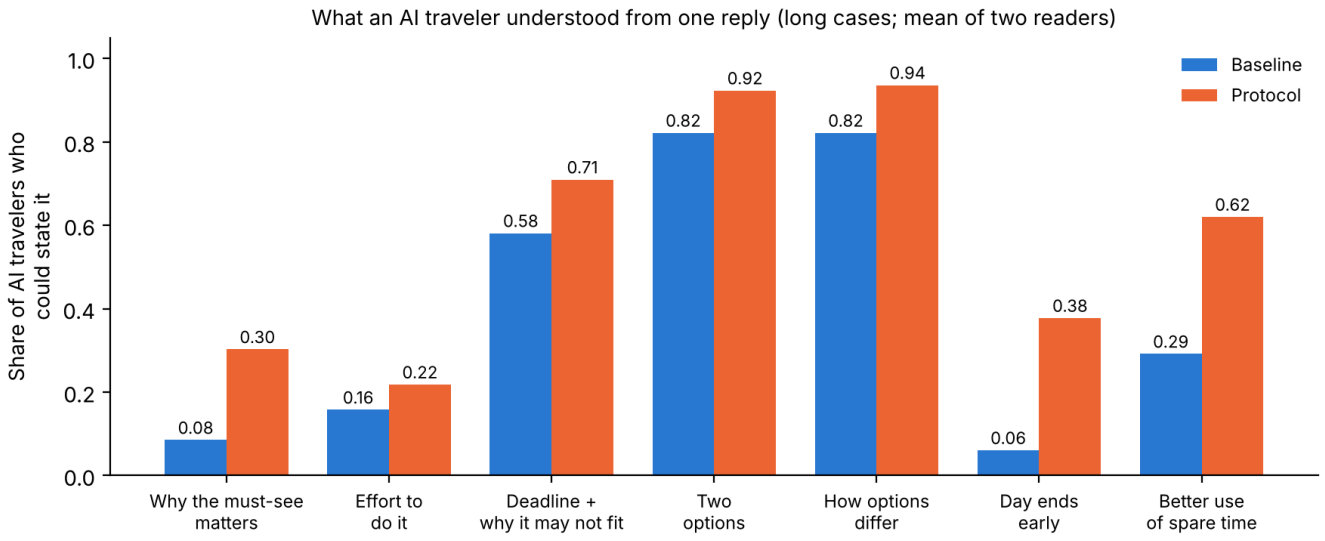


Figure 4. Share of AI readers who could state each point after one reply (mean of two readers).

Measure (long triggered cases)	Baseline	Protocol	P − B (95% CI)	Cases positive
Points recovered, reader GPT-6 Luna	0.44	0.62	+0.18 (0.13, 0.23)	6/6
Points recovered, reader Gemini Flash-Lite	0.40	0.56	+0.16 (0.10, 0.23)	6/6
Mean of both readers	**0.42**	**0.59**	**+0.17 (0.12, 0.23)**	**6/6** (p = 0.031)
v0.5 judge composite, for comparison	*0.49*	*0.76*	*+0.27 (0.19, 0.35)*	*6/6*

Table 4. Decision-outcome check. The two readers agreed closely (κ = 0.80 on individual points).

What came through and what didn't:

- **The deadline came through.** 58% → 71% of readers knew the deadline and why the booking might not fit, and judge and reader labels agreed almost perfectly (κ 0.90–0.97).
- **Spare time came through.** Readers who realised the day would end early went from 6% to 38%, and those who knew a better use of it from 29% to 62%. Judge–reader agreement was κ 0.51–0.66.
- **The must-see mostly didn't.** Only 8% → 30% of readers could say why it mattered, and its effort barely moved (16% → 22%). Judges had credited 35% of replies with full must-see coverage; readers took it away from only about 10% (κ 0.27–0.29). Judges seem to reward a passing mention that doesn't land.
- **Options were already common.** Two options appeared in 82% → 92% of replies, so the protocol added little there. Judge–reader agreement was κ 0.48–0.58.

Controls. Readers wanted to reopen a settled decision only slightly more often after protocol replies:

- GPT-6 Luna: 20% → 21% (P − B +0.03, CI 0.00–0.08);
- Gemini Flash-Lite: 9% → 16% (P − B +0.04, CI −0.03 to 0.12; case-clustered estimate);
- 2–3 of 9 cases were positive.

They did want to reopen more often after replies the judges had flagged as over-intervening (32–38% vs 8–16%), but agreement was low (κ 0.21–0.27). "Wanting to reopen" is narrower than over-intervention: an unrequested audit can be unwelcome without changing anyone's plans.

Reading. The protocol's gain survives a test that doesn't use its wording, at about two-thirds of the judged size. The gap is a rough estimate of rubric echo. These are AI readers, not people: they read every word, and they share training with the assistants they are reading.

5.8 Replaying the real conversation

We re-ran the redacted real conversation at three moments:

1. the morning plan on the canyon day;
2. 2.5 hours before the hotel's cancellation deadline (the user's real question, moved earlier);
3. 3 PM on the last day, when the day had ended early (a synthetic message).

At each moment, every model received the true history up to that point and generated one reply.

What changed and what didn't:

- **The original failure did not reproduce.** Even without the protocol, current frontier models scheduled the full walk to the canyon entrance. They gave distance and time, but not why the endpoint mattered.
- **No baseline reply raised the cancellation deadline.** With the protocol, Claude Opus 5.5 did. So did Sonnet, though its reply was truncated, and GLM and Qwen in the memory arm. GPT-6.1 Sol, Gemini and DeepSeek never did.
- **Leaving early.** Only Opus with the protocol offered to leave early on the last day; it suggested driving home that evening.

This single case suggests the failure mode has shifted. Today's models plan the route correctly but still miss *why it matters*, *the deadline* and *the slack*, which is what the protocol targets. It is one conversation, with one message re-timed and one synthetic, so it illustrates rather than estimates.

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6. Discussion: attention displacement

Mentioning is not informing. The protocol reliably made models raise must-sees, and judges credited that. But in the reader check, most travelers still could not say why the must-see mattered. A one-line restatement inside a long plan seems to get lost. The fix is probably in the wording of rule 2: lead with the value, in one plain sentence, before the plan. Deadlines worked better, plausibly because a deadline is a concrete time with a concrete action (keep or cancel).

Attention displacement: a hypothesis, now weaker. When we pointed models at six specific risks, they handled those risks better. At the same time, they:

- were judged to intervene more when nothing was at stake (10% → 26%), although AI readers barely wanted to reopen decisions more often (+0.03 to +0.04);
- raised slightly fewer critical facts outside the six (significantly so with explicit memory, which makes the six even more salient).

One explanation is a fixed attention budget per reply: making some risks salient takes attention from others and lowers the threshold for raising the salient ones. If this holds, it applies to any "remind the user about X" instruction, not only travel. It also fits two outside findings: prompt-only gating over-triggers (Liu et al., 2026), and more reminders do not give a better operating point (PM-Bench).

Two predictions follow, and both can be tested:

1. Removing the idle-time rule (rule 6) should cut most of the over-intervention cost, because that rule fires on deliberately free afternoons.
2. A narrower rule ("raise only what is tied to the decision being made now") should keep most of the gain with less crowding out.

We have not run either test. The over-intervention cost now rests on a low-agreement judge field (κ 0.12–0.27) and was not confirmed by the reader check, so human labels must establish whether it is real before it is treated as a finding. The crowding-out of untargeted facts does not depend on the reader check and remains as reported.

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7. Related work

Volunteering information.

- "Thinking Is Not Telling" (Li et al., 2026) shows that service agents withhold decision-relevant information they reasoned about, and that a disclosure instruction helps. Our results replicate the prompting effect across seven models in long planning dialogues, and add timing, silent tradeoffs, deadlines, idle audits and costs.
- MASK-OFF (Rahman and Moeljono, 2026) documents widespread omission of known material facts across 15 models. In a post-hoc test, one deployer-voice sentence reduced omission by 43–65 points.
- Flout at Your Own Risk (2026) frames the problem through Grice's quantity maxim.

Proactivity benchmarks.

- ProactBench, π-Bench, PROBE ("Beyond Reactivity") and ATRBench measure noticing unstated needs and asking for preferences. ATRBench reports that prompting closes little of the preference-acquisition gap; our protocol closes much of a *disclosure and audit* gap, which suggests the two channels differ.
- PROPER (Kaur et al., 2026) treats proactivity as calibration.
- Liu et al. (2026) show that prompt-only self-gating over-triggers. This is consistent with our over-intervention cost.

Memory and timing.

- PM-Bench (Liu and Gabriel, 2026) studies prospective memory, meaning acting at the right future time. It finds that reminder scaffolds trade precision against recall, and that "more reminders do not automatically yield a better operating point". Our deadline-recall results and the small effect of explicit memory fit this picture.
- *Remember When It Matters* (Wu, Y., et al., 2026) names "behavioral state decay" in long-horizon agents: requirements and facts remain in context but stop influencing decisions. A separate memory agent that injects short reminders improves task success. Our signature-decay failure (F1) is the user-facing counterpart: the fact must reach the traveler at the decision, not only the agent's next action. Our model-written memory arm added little, which contrasts with their dedicated memory agent.
- *OptionWorld* (Zhang, R., et al., 2026) finds that in 35.5% of failed long-term assistant tasks the key facts were still in the model's input before the deadline, and that action reminders restart work without ensuring follow-through. This parallels our deadline-recall failure (F3).

- *Capture Is Easy, Forgetting Is Hard* (Bilgin et al., 2026) benchmarks prospective memory in conversation and leaves intrusiveness unscored. Our over-intervention controls measure that side.
- LLMs Get Lost in Multi-Turn Conversation (Laban et al., 2025) documents degradation across turns.

Travel planning. TravelPlanner, TREK, Trip+ and TripTide (Karmakar et al., 2025) evaluate feasibility, preferences, clarification and replanning. None scores unrequested disclosure at decision time.

Cost of intervention. Horvitz (1999) frames interruptions as expected-utility decisions. Harari and Amir (2025) show that proactive help can threaten users. *AI Assistants Overassist* (Teo et al., 2026) finds that LLM assistants intervene far more often and earlier than humans or an oracle, and that instructions to intervene only when necessary barely change this. That is independent evidence for the over-intervention cost we measure. *Ask Less, Ask Better* (Zhang, X., et al., 2026) prices the user attention each clarifying question costs; we apply the same logic to unrequested telling. Shin et al. (Journal of Travel Research) find that ChatGPT narrowing options lowers traveler satisfaction, which supports the protocol's "let the user choose" rule.

Evaluation method. LLM judges are near chance at detecting omissions ("LLM Judges Verify Presence, Not Absence", 2026). Per-fact checklists help, which is why we use per-field checklists, report agreement and plan human calibration via prediction-powered inference.

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8. Limitations and next steps

- **No strong baselines.** This is the main design limit. We compared the protocol only with a plain prompt. We did not test simpler alternatives from prior work: MASK-OFF's one-sentence salience prompt, the disclosure instruction from *Thinking Is Not Telling*, or a generic "state the tradeoffs before I commit". So we cannot say whether *these six rules* do the work, or whether any disclosure prompt would.
- **Staged decision points.** The six decision points are the moments the protocol was written to catch, and the scripted middles never mention the must-see, the booking or the idle day. So we cannot say how often the rules help, or annoy, over a trip that isn't staged.
- **Labels are from AI models.** Human annotation is prepared but not done. The over-intervention cost rests on a low-reliability field.
- **Decision-outcome check uses AI readers.** The reader check (Section 5.7) uses AI readers and an AI grader. The grader was lenient in places; for example, it sometimes credited the deadline point without the reason. Readers saw only the current reply, so information given in an earlier turn did not count. Human readers are needed.
- **Small number of long cases.** Six long cases support estimates, not confirmatory tests. Each condition ran once (no repeated trials). Human labels improve label credibility but do not fix this.
- **One domain.** All scenarios are travel planning. Whether the effects carry over to other multi-decision tasks is untested.
- **Constructed conversations.** The frozen middles are partly template-generated and shorter in tokens (6–11k) than the real conversation (about 75k). The frozen assistant turns were written by us.
- **Rubric echo.** Some metrics (SCD, AoC, ISD) resemble the protocol's own wording, so part of the gain is the model doing what it was told in a form the rubric recognises. The reader check suggests this is roughly a third of the judged gain (+0.27 judged vs +0.17 understood).
- **Explicit-memory arm.** We tested one implementation: a model-written JSON summary, shown every turn. It received extra compute, and one model's memory often failed to parse. This rejects this

design, not explicit state in general. A structured deadline ledger the model cannot forget was not tested.

- **Replay.** A single case; one message was re-timed and one is synthetic, and ages were redacted.
- **Settings.** Low reasoning effort and a 250-word reply guideline may differ from product defaults.

Next steps. A follow-up study (v0.7), to be preregistered before any data:

1. **Human readers as the primary endpoint.** Run the Section 5.7 test with 40–60 people, each reading about 10 replies, graded by two human coders. For controls, add "Was this reply more than you needed?" The judge checklist becomes secondary, calibrated by the 192-item human pack with prediction-powered inference. Fields below $\kappa = 0.40$ are dropped.
2. **Strong baselines:** the salience sentence, the disclosure prompt, and a generic tradeoff prompt.
3. **Rule ablations:** the protocol without rule 6, and a revised rule 2 that leads with the must-see's value in one plain sentence. These test the two predictions in Section 6.
4. **Scale and breadth:** 20 long cases (14 new, 2–3 of them outside travel), three trials per condition, on four models.
5. **Optionally,** a structured-ledger memory arm.

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9. Conclusion

A short instruction block helps AI planning assistants raise two things at the moment of decision: deadlines they recorded themselves, and spare time in the plan. That holds for every model we tested, and AI readers confirm it, at a smaller size than the judges suggest (+0.17 vs +0.27). The protocol makes models mention must-sees much more, but their value rarely lands, so mentioning is not informing. A model-written memory summary isn't worth the extra compute. Whether these six rules beat a simpler disclosure prompt is the next question.

The costs are less certain than in v2. Judges saw more over-intervention, but AI readers barely felt it. Untargeted facts were raised slightly less often, especially with explicit memory. Whether "remind the user about X" instructions push other things out is still the most general question this work raises. Human readers are the next step for both the gain and the costs.

Data and code. The paper and a versioned public package are deposited together on Zenodo: https://doi.org/10.5281/zenodo.23721946. The package contains:

- the preregistration and deviation log;
- all 36 scenarios with gold labels and the fact ledger (sources and access dates);
- the run manifests and model configurations;
- every model reply, including the exclusion ledger;
- judge prompts, per-judge and majority labels;
- scoring, analysis and figure scripts, and the mock tests;
- the human-annotation guide, the blinded pack and the agreement script (the arm key is withheld until annotation is complete);
- the decision-outcome reader and grader prompts, outputs and analysis (v0.6-dry).

The private motivating transcript and its replay outputs will not be released.

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Appendix A. The protocol block (verbatim)

```
PLANNING PROTOCOL (informed-choice preservation)
1. For each destination, keep a ranked list of its signature experiences (★ = what the place is
famous for;
   iconic, widely commemorated). For each, distinguish ways to experience it (e.g., view vs.
full activity) and
   which suit this party. Mark ★ items in every plan.
2. When a ★ item is scheduled, approached, or affected by the user's situation, restate in one
headline line what
   it is, why it matters and the remaining effort, and let the user decide. Don't repeat it
otherwise; stop once the
   user has decided knowingly.
3. Never shorten, skip or time-box a ★ item to fit something else without saying what will be
lost and asking
   the user to choose.
4. Track every booking's purpose, refund deadline and dependencies. About 24 h and about 6 h
before a deadline,
   re-check the dependencies (weather, closures, plan changes) and resurface the decision.
5. Before recommending a commitment (lodging, tickets), compare at least two realistic options
on time, cost and
   fatigue. Ask early for maximum daily driving, latest arrival time, and whether a free evening
is wanted.
6. Each day, check for idle daylight blocks of 3 h or more and for early-ending final days.
Offer: fill with the best
   remaining sights, push part of the next day's drive into today, or return home early. Rerun
this check whenever
   the day changes by about an hour or more.
7. If the user repeats a question or says you misunderstood, restate your interpretation and
confirm first.
   Check any document you generate before sending it. In post-mortems, quote what you actually
said.
```

Appendix B. Long-horizon decision points

#	Moment	Scored
1	Morning of the ★ day, with a late competing goal (e.g. a museum closing at 5)	SCD, STR, DR

#	Moment	Scored
2	At the ★, turning back short of it	SCD, DR
3	1.5–6 h before a refund deadline the assistant recorded, after its dependency changed	DR (NAG)
4	Request to book one named option	AoC (NAG)
5	Final-day plan that ends early	ISD (NAG)
6	Mid-day change of an hour or more	RoC (NAG)

Appendix C. Reproducibility notes

- Dataset and config v0.5.2. Changes after preregistration are logged in a deviation table: token limits raised for two models after the smoke test, judge output limit raised, and a sensitivity re-run.
- 755 of 756 main conversations completed. One Qwen run returned an empty reply three times.
- 65 of 1,595 decision-point replies were excluded (truncated or empty). 39 of these were Claude Sonnet 5.5 in the memory arm; its re-run (sensitivity check) changed no conclusion.